Established – 1961 Subject: **WEB PROGRAM**

SEVA SADAN'S

R. K. TALREJA COLLEGE

OF

ARTS, SCIENCE & COMMERCE

ULHASNAGAR – 421 003

CERTIFICATE

This is to certify that Mr./Ms. SANJEEV SHARMA of F.Y. Information Technology (FYIT) Roll No. _2541048_ has satisfactorily completed the Web Designing Mini Project entitled Portfolio with Performance Optimized Media Handling during the academic year 2025 – 2026, as a part of the practical requirement. The project work is found to be satisfactory and is approved for submission.

PROF. INCHARGE **HEAD OF DEPT**

SAHIL SHUKLA **LAXMI JESWANI**

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INTRODUCTION:

This project is a modern, responsive **Personal Portfolio Website** designed to serve as a high-impact digital identity for **Chirag Bhakuni**, an aspiring IT professional and FY BSc.IT student. Built with a focus on clean aesthetics and user experience, the site features a sophisticated **dark-mode glassmorphism** design, utilizing a deep gradient palette and sleek typography. Key technical highlights include **dynamic JavaScript greetings** that adapt to the user’s local time, smooth entry animations powered by **ScrollReveal**, and a fully fluid layout that ensures a seamless viewing experience across both desktop and mobile devices. More than just a resume, this platform integrates social connectivity and interactive UI elements to bridge the gap between technical skill and creative personality.".

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REQUIREMENT SPECIFICATION :

"The technical requirement for this portfolio was to develop a **high-performance, single-page digital identity** that balances aesthetic appeal with functional programming. Architecturally, the system is required to implement a **responsive layout** using CSS Flexbox, ensuring that the 'Z-pattern' visual hierarchy—where the user's eye moves from the profile branding to the navigation and then to the call-to-action—remains intact across all device resolutions. A core functional requirement is the **integration of a real-time logic layer** via JavaScript, which fetches the user's system clock to deliver a personalized, time-sensitive greeting. Furthermore, the specification mandates the use of **external API libraries** (FontAwesome and ScrollReveal) to enhance the UI/UX without sacrificing load speeds, utilizing 'lazy loading' for media assets to optimize performance. The final deliverable serves the dual requirement of a professional contact directory and a creative showcase, wrapped in a **glassmorphism design language** that emphasizes depth, transparency, and modern interactivity."

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SYSTEM DESIGN:

1. Architectural Overview

The system is built using a **Modular Front-End Design**. It doesn't require a database (it is "stateless"), making it incredibly fast and secure. The architecture can be divided into three distinct layers:

- **The Content Layer (HTML5):** This acts as the skeleton of the system. It organizes information into semantic sections: `<header>`, `<nav>`, and `<section class="hero">`.
- **The Presentation Layer (CSS3):** This layer handles the "Glassmorphism" aesthetic. It uses a **Flexbox-based Grid System** to ensure the layout is fluid and responsive.
- **The Behavioral Layer (JavaScript):** This acts as the "brain." It interacts with the **Document Object Model (DOM)** to change text based on time and triggers external API libraries for animations.

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2. Component Breakdown

A. Navigation & Branding Module

This component is designed to be "persistent."

- **Logic:** Uses `position: fixed` and `z-index: 1000` to stay above all other content.
- **UX Design:** Uses `backdrop-filter: blur` to maintain readability while scrolling over background elements.

B. Dynamic Greeting Engine (The Script)

The system contains a logical function called `setGreeting()`.

1. **Input:** It pulls the current system time from the client's machine.
2. **Processing:** It runs a conditional "if-else" check against the `hour` variable.
3. **Output:** It injects specific strings (e.g., "Good Morning") into the HTML via `innerHTML`.

C. Animation & Motion Controller

Instead of heavy video files, the system uses two lightweight methods for motion:

- **CSS Keyframes:** Handles the infinite "pulse" effect on the hero image background.

- **ScrollReveal.js:** An external module that listens for "scroll events" and reveals hidden elements with a 60px upward slide.

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3. Data & Resource Flow

The website follows a specific sequence when a user visits the URL:

1. **Request:** Browser requests the `index.html`.
2. **External Calls:** The system fetches **Google Fonts** and **FontAwesome Icons** from their respective Content Delivery Networks (CDNs).
3. **Render:** The browser builds the DOM and applies CSS rules.
4. **Execution:** The JavaScript runs immediately after the DOM is loaded to set the greeting and initialize the scroll observers.

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4. Design Patterns Used

- **Responsive Web Design (RWD):** Uses a "Mobile-First" mindset for the breakpoint at 900px.
- **Single Page Application (SPA) Style:** All main content is on one page to reduce loading times between clicks.

- **Z-Pattern Layout:** Matches the natural human scanning pattern (Top-left branding $\rightarrow$ Top-right links $\rightarrow$ Center content).

SYSTEM IMPLEMENTATION:

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
   <title>My Portfolio</title>
   <link
href="https://fonts.googleapis.com/css2?famil
y=Poppins:wght@300;400;600;700&display=
swap" rel="stylesheet">
   <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/f
ont-awesome/6.4.0/css/all.min.css">
   <script
src="https://unpkg.com/scrollreveal"></scrip
t>

   <style>
      *{ margin:0; padding:0; box-
sizing:border-box; font-family:'Poppins', sans-
serif; }

      body{
         background: linear-
gradient(135deg,#0f2027,#1f4037,#2c5364);
```

```
    color:white;
    overflow-x:hidden;
}

header{
    display:flex;
    justify-content:space-between;
    align-items:center;
    padding:15px 8%;
    position: fixed;
    top: 0; width: 100%; z-index: 1000;
    background: rgba(15, 32, 39, 0.85);
    backdrop-filter: blur(10px);
}

/* --- LOGO / PROFILE BOX --- */
.profile-box-nav {
    display: flex;
    align-items: center;
    gap: 12px;
}

.profile-pic-nav {
    width: 45px;
    height: 45px;
    border-radius: 50%;
    object-fit: cover;
    border: 2px solid #a8e063;
}
```

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```
.profile-text-nav h4 {
    font-size: 14px;
    font-weight: 700;
    letter-spacing: 0.5px;
    margin: 0;
    color: #fff;
}

.profile-text-nav small {
    font-size: 10px;
    color: #a8e063;
    display: block;
    text-transform: uppercase;
}

/* --- NAVIGATION LINKS (Moved to
Right with More Space) --- */
nav {
    margin-left: auto; /* Pushes nav to the
right */
}

nav ul{
    display:flex;
    list-style:none;
    gap: 50px; /* Increased space between
links */
    padding-right: 20px;
```

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```
}

nav ul li a{
   text-decoration:none;
   color:white;
   font-size:15px;
   font-weight: 400;
   transition:0.3s;
}

nav ul li a:hover{ color:#a8e063; }

.hero{
   display:flex; justify-content:space-
between; align-items:center;
   padding:150px 8% 100px; min-height:
100vh;
}

.hero-left{ max-width:500px; }
.hero-left h5{ font-size:12px; letter-
spacing:2px; margin-bottom:15px; color:
#a8e063; }
.hero-left h1{ font-size:45px; margin-
bottom:10px; }
.hero-left h1 span{ color:#a8e063; }
.hero-left p{ font-size:15px; margin-
bottom:30px; opacity:0.8; }
```

```
.buttons{ display:flex; gap:20px; margin-bottom:25px; }
.btn1{ background:#a8e063; color:#0f2027; padding:10px 22px; border-radius:25px; text-decoration:none; font-weight: 600; }

.social a{
    color: white; background: rgba(255,255,255,0.1);
    width: 35px; height: 35px; display: inline-flex;
    align-items: center; justify-content: center;
    border-radius: 50%; margin-right: 10px; transition: 0.3s;
}
.social a:hover{ background: #a8e063; color: #0f2027; transform: translateY(-3px); }

/* Circle Image Logic for Hero Right */
.hero-right{ position:relative; }
.hero-right::before{
    content:""; position:absolute; width:400px; height:400px;
    background: radial-gradient(circle,#a8e06344,transparent);
    border-radius:50%; z-index:0; animation:pulse 3s infinite alternate;
```

```
        top: 50%; left: 50%; transform:
translate(-50%, -50%);
    }

    .hero-right img{
        width:350px; height:350px; border-
radius:50%;
        object-fit: cover; position:relative; z-
index:2;
        border: 10px solid rgba(168, 224, 99,
0.1);
        box-shadow: 0 20px 50px
rgba(0,0,0,0.3);
        transition: 0.4s ease;
    }

    .hero-right img:hover{ transform:
scale(1.03); border-color: rgba(168, 224, 99,
0.3); }

    @keyframes pulse{
        from{ opacity:0.5; transform: translate(-
50%, -50%) scale(1); }
        to{ opacity:0.8; transform: translate(-
50%, -50%) scale(1.1); }
    }

    @media(max-width:900px){
        .hero{ flex-direction:column-reverse;
```

```
text-align:center; }
        nav ul{ display:none; }
        .hero-right{ margin-bottom:50px; }
        .hero-right img{ width: 280px; height:
280px; }
    }
  </style>
</head>

<body>

<header>
  <div class="profile-box-nav">
    <img
src="C:\Users\admin\Downloads\chirag
project\3e2db926-cc27-4554-a090-
fcb3003b4e07.jpg" alt="Logo" class="profile-
pic-nav">
    <div class="profile-text-nav">
      <h4>CHIRAG BHAKUNI</h4>
      <small>FY BSc.IT STUDENT</small>
    </div>
  </div>

  <nav>
    <ul>
      <li><a href="#">Home</a></li>
      <li><a href="#">Services</a></li>
      <li><a href="#">Projects</a></li>
```

```
            <li><a href="#">Contact</a></li>
        </ul>
    </nav>
</header>

<section class="hero">
    <div class="hero-left">
        <h5 class="reveal"> More than a portfolio,
it's my Digital identity □</h5>
        <h1 class="reveal" id="greetText">hi,
<span>Chirag here</span></h1>
        <h2
class="reveal">learner/Editor/Gamer</h2>
        <p class="reveal">Building / Learning
Experiences beyond limits.</p>
        <div class="buttons reveal">
            <a href="#" class="btn1">About
Me</a>
        </div>
        <div class="social reveal">
            <a
href="https://www.instagram.com/yaa.itz.chir
uuu?igsh=MWU4eWNsdjFtOTQ1eg=="
target="_blank" title="Instagram">
                <i class="fab fa-instagram"></i>
            </a>
            <a href="https://wa.me/919321427975"
target="_blank" title="WhatsApp">
                <i class="fab fa-whatsapp"></i>
```

```
            </a>
            <a
href="mailto:chiragbhakuni07@gmail.com"
title="Email">
                <i class="far fa-envelope"></i>
            </a>
        </div>
    </div>

    <div class="hero-right reveal">
        <img
src="C:\Users\admin\Downloads\chirag
project\3e2db926-cc27-4554-a090-
fcb3003b4e07.jpg" alt="SS" loading="lazy">
    </div>
</section>

<script>
    function setGreeting() {
        const hour = new Date().getHours();
        let greeting = "";
        if (hour < 12) greeting = "Good Morning ☀
";
        else if (hour < 17) greeting = "Good
Afternoon □";
        else if (hour < 21) greeting = "Good
Evening □";
        else greeting = "Good Night □";
```

```
document.getElementById("greetText").inner
HTML = `HEY, ${greeting}
<br><span>CHIRAG HERE !</span>`;
    }
    setGreeting();

    // ScrollReveal for Lazy Animations
    ScrollReveal().reveal('.reveal', {
        distance: '60px',
        duration: 1200,
        easing: 'cubic-bezier(0.5, 0, 0, 1)',
        origin: 'bottom',
        interval: 150
    });
</script>

</body>
</html>
```

SYSTEM TESTING AND RESULT:

1. System Testing Phase

Testing for this portfolio was conducted using a **Black Box Testing** methodology, where the focus was on inputs (clicks, screen resizing) and outputs (animations, layout changes) without modifying the internal code during the test.

2. Test Environment

- **Operating System:** Windows 10/11
- **Browsers:** Google Chrome (v120+), Microsoft Edge, and Safari (iOS)
- **Resolution Testing:** 1920x1080 (Desktop), 768x1024 (Tablet), and 390x844 (Mobile)

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3. Results and Conclusion

Visual Result

The final output is a high-fidelity, interactive interface. The **Glassmorphism** effect remains consistent across different backgrounds, and the typography remains legible. The use of the `backdrop-filter` creates a professional, layered depth between the navigation bar and the hero content.

RESULT:

https://sanjiv0069.github.io/my-portfolio

FUTURE SCOPE AND CONCLUSION:

While the current version of the portfolio is a functional and aesthetic success, there are several technical enhancements planned for future iterations:

- **Backend Integration:** Implementing a **Node.js or PHP backend** to handle a "Contact Me" form, allowing messages to be saved to a database (MySQL/MongoDB) rather than just using a `mailto:` link.
- **Project Gallery with Filtering:** Adding a dynamic "Projects" section using **JavaScript arrays** or a JSON API to allow users to filter work by category (e.g., Web Dev, Editing, Gaming).
- **Dark/Light Mode Toggle:** Adding a theme switcher that allows users to toggle between the current dark aesthetic and a professional light mode.
- **Progressive Web App (PWA) Capabilities:** Converting the site into a PWA so it can be installed on mobile devices and accessed offline.

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- **Performance Optimization:** Hosting images on a Cloudinary or AWS S3 bucket instead of local storage to improve global load times and reliability.

Conclusion

The development of this personal portfolio project has been a comprehensive exercise in modern front-end engineering. By integrating **HTML5**, **CSS3**, and **JavaScript**, I have successfully created a platform that is not only a visual representation of my digital identity but also a demonstration of technical proficiency.

The project highlights the importance of **Responsive Web Design**, ensuring that the user experience is high-quality regardless of the hardware being used. The use of dynamic scripts for time-based greetings and third-party libraries for motion effects shows a transition from static web pages to interactive web applications.

Ultimately, this portfolio serves as a solid foundation for my journey in the IT industry. It

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fulfills the academic requirements of the BSc.IT curriculum while providing a real-world tool for professional networking and personal branding.

REFERENCE:

Visual Studio Code – Used for writing, editing, and managing the website source code.

Gemini AI – Used for idea generation, concept understanding, and guidance during development.

Google Chrome – Used for running, testing, and debugging the portfolio website.

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GLOSSARY:

HTML (HyperText Markup Language):
A standard markup language used to create the structure of web pages.

CSS (Cascading Style Sheets):
A styling language used to design and format the layout of web pages.

JavaScript:
A scripting language used to add interactivity and dynamic behavior to websites.

Responsive Design:
A design approach that ensures the website works properly on different screen sizes like mobile, tablet, and desktop.

CSS Variables:
Reusable custom properties used to store values like colors and make theme management easier.

Theme Toggle:
A feature that allows switching between dark mode and light mode.

Intersection Observer:
A JavaScript API used to detect when elements become

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visible in the viewport.

Canvas API:
A JavaScript feature used to draw graphics and animations directly in the browser.

Animation:
A visual effect that creates movement or transition on web elements.

User Interface (UI):
The visual layout and design through which users interact with the website.

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